DIGITAL ELECTRONICS

Unit-IV: Sequential Circuits

Comprehensive Study Notes

HPU BCA

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1. Introduction to Sequential Circuits

1.1 What are Sequential Circuits?

A **sequential circuit** is a type of digital circuit where the **output depends not only on the current inputs** but also on the **previous state** (past inputs). Unlike combinational circuits, sequential circuits have **memory elements** that store information about past inputs.

```
BLOCK DIAGRAM OF A SEQUENTIAL CIRCUIT

Inputs ---> [COMBINATIONAL LOGIC] ---> Outputs
                   |
                   v
            [MEMORY (Flip-flops)]
                   |
            (Feedback path)
```

1.2 Types of Sequential Circuits

Type	Description	Clock Signal	Examples
Synchronous	Outputs change only on clock edges	Required	Flip-flops, Counters, Shift Registers
Asynchronous	Outputs change immediately when inputs change	Not required	Latches, Asynchronous Counters

1.3 Sequential vs Combinational Circuits

Feature	Combinational Circuit	Sequential Circuit
Memory	No memory	Has memory elements

Output	Depends only on current inputs	Depends on current inputs + previous state
Clock	Not required	Required (synchronous)
Feedback	No feedback	Has feedback paths
Examples	Adder, Decoder, MUX	Flip-flops, Counters, Shift Registers
Complexity	Less complex	More complex

2. Latches

2.1 What is a Latch?

A **latch** is a basic memory element that can store one bit of information. Latches are **level-sensitive** devices — they respond to input signals as long as the enable signal is active.

2.2 SR Latch (Set-Reset Latch)

Circuit using NAND Gates

```
S' ---|NAND|
         |    |---- Q
       --|    |
      |
      |--
      |  |
R' ---|NAND|
         |    |---- Q'
       --|    |
      |
      |--
```

Truth Table (SR Latch with NAND)

S'	R'	Q	Q'	State
0	0	1	1	Invalid
0	1	1	0	Set
1	0	0	1	Reset
1	1	Q_{prev}	Q'_{prev}	No Change

2.3 D Latch (Data Latch)

The **D latch** has a single data input (D) and an enable signal. It eliminates the invalid state of the SR latch.

Truth Table

Enable	D	Q	State
0	X	Q_{prev}	No Change
1	0	0	Reset
1	1	1	Set

3. Flip-Flops

3.1 What is a Flip-Flop?

A **Flip-Flop** is a basic memory element that can store one bit of information. Unlike latches, flip-flops are **edge-triggered** — they change state only on clock edges (rising or falling).

```
Inputs ---> [FLIP-FLOP] ---> Outputs (Q, Q')
Clock --->
```

3.2 Types of Flip-Flops

Type	Inputs	Characteristic Equation	State Changes
SR	S, R	$Q_{next} = S + R'Q$	Set, Reset, No Change, Invalid
JK	J, K	$Q_{next} = JQ' + K'Q$	Set, Reset, No Change, Toggle
D	D	$Q_{next} = D$	Set or Reset
T	T	$Q_{next} = T \oplus Q$	No Change, Toggle

3.3 SR Flip-Flop (Set-Reset)

Truth Table

S	R	Q_{next}	State
0	0	Q (Previous)	No Change
0	1	0	Reset
1	0	1	Set
1	1	-	Invalid (Undefined)

Characteristic Equation

```
Q(next) = S + R'Q
Constraint: S · R = 0 (cannot both be 1)
```

Excitation Table

$Q \rightarrow Q_{next}$	S	R
0 → 0	0	X
0 → 1	1	0
1 → 0	0	1
1 → 1	X	0

3.4 JK Flip-Flop

The **JK flip-flop** is an **improved version** of the SR flip-flop that eliminates the invalid state condition. When J=1 and K=1, the output **toggles**.

Truth Table

J	K	Q_{next}	State
0	0	Q (Previous)	No Change
0	1	0	Reset
1	0	1	Set
1	1	Q' (Toggle)	Toggle

Characteristic Equation

```
Q(next) = JQ' + K'Q
```

Excitation Table

$Q \rightarrow Q_{next}$	J	K
0 → 0	0	X
0 → 1	1	X
1 → 0	X	1
1 → 1	X	0

3.5 D Flip-Flop (Data / Delay)

The **D flip-flop** has **one input (D)** and **one output (Q)**. It transfers the input to the output at the clock edge. It is the simplest flip-flop, commonly used in registers and data storage.

Truth Table

D	Q_{next}	State
0	0	Reset
1	1	Set

Characteristic Equation

```
Q(next) = D
```

Excitation Table

$Q \rightarrow Q_{next}$	D
0 → 0	0
0 → 1	1
1 → 0	0
1 → 1	1

3.6 T Flip-Flop (Toggle)

The **T flip-flop** has **one input (T)**. It toggles the output when T=1 and holds the state when T=0. It is commonly used in counters.

Truth Table

T	Q_{next}	State
0	Q (Previous)	No Change
1	Q' (Toggle)	Toggle

Characteristic Equation

```
Q(next) = T ⊕ Q
```

Excitation Table

Q → Q_{next}	T
0 → 0	0
0 → 1	1
1 → 0	1
1 → 1	0

3.7 Flip-Flop Conversions

Conversion	Required Logic
SR → JK	S = JQ', R = KQ
SR → D	S = D, R = D'
SR → T	S = TQ', R = TQ
JK → D	J = D, K = D'
JK → T	J = T, K = T
D → T	D = T ⊕ Q

4. Edge-triggered vs Level-triggered

Feature	Level-triggered (Latch)	Edge-triggered (Flip-Flop)
Triggering	Responds to input level	Responds at clock edge (rising/falling)
Transparency	Transparent when enabled	Not transparent
Noise Immunity	Less immune	More immune
Usage	Simple storage	Counters, registers, sequential circuits
Race Condition	More prone	Less prone (master-slave eliminates)

Edge Detection Types

```
Positive Edge-Triggered (Rising Edge):
CLK: __|--  -->  Changes occur on upward transition

Negative Edge-Triggered (Falling Edge):
CLK: --|__  -->  Changes occur on downward transition

Level-Triggered (Latch):
CLK: ____  -->  Changes while clock is HIGH (or LOW)
```

5. Master-Slave Flip-Flop

A **Master-Slave flip-flop** consists of **two flip-flops** (master and slave) that operate on different clock phases to eliminate race conditions.

```
MASTER-SLAVE FLIP-FLOP BLOCK DIAGRAM

Inputs ---> [MASTER FF] ---> [SLAVE FF] ---> Outputs
                |
Clock ---------|
                |
Clock' (Complement) ----|
```

Operation

1. **Master** is enabled when clock is HIGH
2. **Slave** is enabled when clock is LOW (inverted)
3. Output changes only on the **falling edge** of the clock
4. This prevents race conditions (multiple changes within one clock pulse)

6. Shift Registers

6.1 What is a Shift Register?

A **Shift Register** is a sequential circuit that can store and **shift** binary data. It is constructed using cascaded flip-flops (typically D flip-flops) with a common clock.

```
Serial In ---> [FF1] ---> [FF2] ---> [FF3] ---> [FF4] ---> Serial Out
                 |          |          |          |
                (Q1)       (Q2)       (Q3)       (Q4)

        Parallel Outputs

All flip-flops share a common CLOCK signal.
```

6.2 Types of Shift Registers

Type	Input	Output	Description
SISO	Serial In	Serial Out	Shift register with serial input and serial output
SIPO	Serial In	Parallel Out	Converts serial data to parallel data
PISO	Parallel In	Serial Out	Converts parallel data to serial data
PIPO	Parallel In	Parallel Out	Stores and shifts parallel data

SISO (Serial In - Serial Out)

```
Data is entered bit by bit serially.
After N clock pulses, the data appears at the serial output.
Used for: Time delays, data transfer.
```

SIPO (Serial In - Parallel Out)

```
Data is entered bit by bit serially.
After N clock pulses, all bits are available in parallel.
Used for: Serial-to-parallel conversion (e.g., receiving serial data).
```

PISO (Parallel In - Serial Out)

```
Data is loaded in parallel (all bits at once).
Data is shifted out bit by bit serially.
Used for: Parallel-to-serial conversion (e.g., transmitting data).
```

PIPO (Parallel In - Parallel Out)

```
Data is loaded in parallel and output in parallel.
Used for: Temporary storage with shift capability.
```

7. Counters

7.1 What is a Counter?

A **counter** is a sequential circuit that goes through a predetermined sequence of states. Counters are used for counting events, frequency division, and timing applications.

7.2 Synchronous Counters

In **synchronous counters**, all flip-flops receive the **same clock signal** simultaneously. They change state at the same time.

- **Advantage:** Faster operation, no propagation delay accumulation
- **Disadvantage:** More complex circuitry
- **Common:** Synchronous up counter, synchronous down counter

3-bit Synchronous Up Counter

```
Clock Pulse | Q2 | Q1 | Q0 | Count
    0       | 0  | 0  | 0  |  0
    1       | 0  | 0  | 1  |  1
    2       | 0  | 1  | 0  |  2
    3       | 0  | 1  | 1  |  3
    4       | 1  | 0  | 0  |  4
    5       | 1  | 0  | 1  |  5
    6       | 1  | 1  | 0  |  6
    7       | 1  | 1  | 1  |  7
    8       | 0  | 0  | 0  |  0 (rollover)
```

7.3 Asynchronous Counters (Ripple Counters)

In **asynchronous counters**, the clock is applied only to the first flip-flop, and the output of each flip-flop serves as the clock for the next.

- **Advantage:** Simple circuit design
- **Disadvantage:** Propagation delay accumulates (ripple effect)
- **Common:** Ripple counter, decade counter

> **Note:** Asynchronous counters are slower than synchronous counters for large bit widths due to cumulative propagation delays.

7.4 Ring Counter

A **ring counter** is a shift register where the output of the last flip-flop is fed back to the input of the first flip-flop, creating a ring. Only one flip-flop is set to 1 at any time.

```
4-bit Ring Counter Sequence:
Clock | Q0 | Q1 | Q2 | Q3
  0   | 1  | 0  | 0  | 0
  1   | 0  | 1  | 0  | 0
  2   | 0  | 0  | 1  | 0
  3   | 0  | 0  | 0  | 1
  4   | 1  | 0  | 0  | 0 (repeats)
```

7.5 Johnson Counter (Twisted Ring Counter)

A **Johnson counter** is a shift register where the complemented output of the last flip-flop is fed back to the input of the first. It produces a sequence of 2n states for n flip-flops.

```
4-bit Johnson Counter Sequence:
Clock | Q0 | Q1 | Q2 | Q3
  0   | 0  | 0  | 0  | 0
  1   | 1  | 0  | 0  | 0
  2   | 1  | 1  | 0  | 0
  3   | 1  | 1  | 1  | 0
  4   | 1  | 1  | 1  | 1
  5   | 0  | 1  | 1  | 1
  6   | 0  | 0  | 1  | 1
  7   | 0  | 0  | 0  | 1
  8   | 0  | 0  | 0  | 0 (repeats)
```

For 4 flip-flops, Johnson counter has **8 distinct states** (2n = 2×4 = 8).

Ring vs Johnson Counter

Feature	Ring Counter	Johnson Counter

Feedback	Q_{last} → D_{first}	Q'_{last} → D_{first}
States	n states	2n states
Usage	Sequencing, timing	Frequency division, decoding

8. Applications of Sequential Circuit

Application	Circuit Type	Description
Data Storage	Registers (Flip-flops)	Registers store binary
Frequency Division	Counters	Counters divide clock f
Serial-to-Parallel Conversion	SIPO Shift Register	Converts serial data st
Parallel-to-Serial Conversion	PISO Shift Register	Converts parallel data
Time Delay	SISO Shift Register	Provides time delay pro
Event Counting	Counters	Counts occurrences of e
Sequence Detection	Finite State Machines	Detects specific sequen
Memory Addressing	Counters	Generates sequential ac
Digital Clocks	Counters + Registers	Keeps time using counte
Debouncing	Flip-flops	Eliminates switch bounc

9. Quick Revision

Flip-Flop Summary

Flip-Flop	Inputs	Characteristic Equation	Usage
SR	S, R	$Q_{next} = S + R'Q$	Basic memory
JK	J, K	$Q_{next} = JQ' + K'Q$	General purpose, counters
D	D	$Q_{next} = D$	Registers, data storage
T	T	$Q_{next} = T \oplus Q$	Counters, toggling

Shift Register Summary

Type	Input	Output	Application
SISO	Serial	Serial	Delay elements
SIPO	Serial	Parallel	Serial-to-parallel conversion
PISO	Parallel	Serial	Parallel-to-serial conversion
PIPO	Parallel	Parallel	Register storage

Counter Summary

Type	Clock	Speed	Circuit
Synchronous	Same clock to all FFs	Fast	More co
Asynchronous (Ripple)	Clock ripples through FFs	Slow (for large n)	Simple
Ring	Same clock to all FFs	Fast	Simple
Johnson	Same clock to all FFs	Fast	Simple

10. HPU Important Questions

Short Answer Questions (2-5 Marks)

1. What is a sequential circuit?
2. What is the difference between combinational and sequential circuits?
3. What is a latch? Differentiate between latch and flip-flop.
4. What is a flip-flop?
5. What is the difference between SR and JK flip-flop?
6. What is a D flip-flop? Write its characteristic equation.
7. What is a T flip-flop? Write its characteristic equation.
8. What is a shift register? Name its types.
9. What is the difference between synchronous and asynchronous counters?
10. What is a ring counter?
11. What is a Johnson counter?
12. What is the difference between a multiplexer and decoder?

Long Answer Questions (10 Marks)

1. Explain the difference between combinational and sequential circuits with e
2. Explain SR, D, JK, and T flip-flops with truth tables, characteristic equat
3. Explain edge-triggered vs level-triggered flip-flops.

4. Explain Master-Slave flip-flop with block diagram and operation.
5. What is a shift register? Explain SISO, SIPO, PISO, and PIPO with diagrams.
6. Explain synchronous and asynchronous counters with examples.
7. Explain ring counter and Johnson counter with state sequences.
8. Explain the applications of sequential circuits in digital systems.
9. Differentiate between synchronous and asynchronous sequential circuits.

Objective Questions (1 Mark)

1. A sequential circuit has: (a) No memory (b) Memory (c) Only inputs (d) None
   Answer: (b)
2. An SR flip-flop has an invalid state when: (a) R=0, S=0 (b) R=0, S=1 (c) R=
   Answer: (d)
3. JK flip-flop eliminates the invalid state of: (a) RS flip-flop (b) D flip-f
   Answer: (a)
4. The characteristic equation of D flip-flop is: (a) Q = D (b) Q = D' (c) Q =
   Answer: (a)
5. The characteristic equation of T flip-flop is: (a) Q = D (b) $Q = T \oplus Q$ (c)
   Answer: (b)
6. A shift register with serial input and parallel output is: (a) SISO (b) SIP
   Answer: (b)
7. A 4-bit ring counter has how many states? (a) 4 (b) 8 (c) 16 (d) 2
   Answer: (a)
8. A 4-bit Johnson counter has how many states? (a) 4 (b) 8 (c) 16 (d) 2
   Answer: (b)
9. In synchronous counters, all flip-flops receive: (a) Different clocks (b) S
   Answer: (b)
10. A latch is: (a) Edge-triggered (b) Level-triggered (c) Both (d) None
    Answer: (b)

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